

BRAC University
MAT-215
Exercise Sheet #1

Part A

1. Perform each of the indicated operations:

$$\begin{array}{lll} \text{(i)} (i-2)\{2(1+i)-3(i-1)\} & \text{(ii)} \frac{(2+i)(3-2i)(1-i)}{(1-i)^2} & \text{(iii)} (2i-1)^2 \left\{ \frac{4}{1-i} + \frac{2-i}{1+i} \right\} \\ \text{(iv)} 3\left(\frac{1+i}{1-i}\right)^2 - 2\left(\frac{1-i}{1+i}\right)^3 & \text{(v)} \frac{3i^{10} - i^{19}}{2i-1} & \text{(vi)} \frac{i^4 - i^9 + i^{16}}{2-i^5 + i^{10} - i^{15}} \end{array}$$

2. Show that (i) $(5+3i) + \{(-1+2i) + (7-5i)\}$ and (ii) $\{(5+3i) + (-1+2i)\} + (7-5i)$ illustrate the associative law of addition.

3. If $z_1 = 1-i$, $z_2 = -2+4i$, $z_3 = \sqrt{3}-2i$, evaluate each of the following:

$$\begin{array}{lll} \text{(i)} |2z_2 - 3z_1|^2 & \text{(ii)} \left| \frac{z_1 + z_2 + 1}{z_1 - z_2 + i} \right| & \text{(iii)} \overline{(z_2 + z_3)(z_1 - z_3)} \quad \text{(iv)} \operatorname{Re}\{2z_1^3 + 3z_2^3 - 5z_3^2\} \\ \text{(v)} \operatorname{Im}\left\{ \frac{z_1 z_2}{z_3} \right\} & \text{(vi)} z_1^2 + 2z_1 - 3 & \text{(vii)} |z_1 \overline{z_2} + z_2 \overline{z_1}| \quad \text{(viii)} \frac{1}{2} \left(\frac{z_3}{z_3} + \frac{\overline{z_3}}{z_3} \right) \\ \text{(ix)} (z_3 - \overline{z_3})^5 \end{array}$$

4. Express each of the following complex number in polar form and show them graphically.

$$\text{(i)} 2 + 2\sqrt{3}i \quad \text{(ii)} 2\sqrt{2} + 2\sqrt{2}i \quad \text{(iii)} -2\sqrt{3} - 2i \quad \text{(iv)} -1 + \sqrt{3}i \quad \text{(v)} -\sqrt{6} - \sqrt{2}i$$

5. Prove that: (i) $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$ (ii) $|z_1 z_2| = |z_1| |z_2|$ (iii) $\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$
(iv) $|z| \geq \operatorname{Re}(z)$.

6. (i) State **De Moivre's Theorem**

(ii) Apply **De Moivre's Theorem** to find $(1+i)^{20}$

7. Evaluate each of the following by **De Moivre's Theorem**:

$$\begin{array}{lll} \text{(i)} \frac{(8\operatorname{cis}40^\circ)^3}{(2\operatorname{cis}60^\circ)^4} & \text{(ii)} \frac{(3e^{\frac{\pi}{6}})(2e^{\frac{-5\pi}{4}})(6e^{\frac{5\pi}{3}})}{(4e^{\frac{2\pi}{3}})^2} & \text{(iii)} (5\operatorname{cis}20^\circ)(3\operatorname{cis}40^\circ) \quad \text{(iv)} (2\operatorname{cis}50^\circ)^6 \end{array}$$

8. Find all the roots of the following equations.

(i) $(-1+i)^{\frac{1}{3}}$ (ii) $z^5 = -4 + 4i$ (iii) $z^4 = -16i$ (iv) $z^6 = 64$ (v) $z^4 + z^2 + 1 = 0$.

(vi) $(-4 + 4i)^{1/5}$ (vii) $(-2\sqrt{3} - 2i)^{1/4}$

Part B

1. Perform the indicated operations analytically and graphically.

(a) $(2 + 3i) + (4 - 5i)$ (b) $(7 + i) - (4 - 2i)$

2. Describe geometrically the set of points z satisfying the following conditions:

(a) $\operatorname{Re}(z) > 1$

(b) $|2z + 3| > 4$

(c) $\operatorname{Re}\left(\frac{1}{z}\right) > 1$

(d) $1 < |z - 2i| < 2$

(e) $|z + 1 - i| \leq |z - 1 + i|$

(f) $\operatorname{Re}(z) \geq 0$

(g) $|z - 4| \geq |z|$

(h) $|z - 2| \leq |z + 2|$

(i) $\operatorname{Re}(1/z) \leq 1/2$

(j) $\pi/2 < \arg z < 3\pi/2, |z| > 2$

3. Using the properties of conjugate and modulus, show that:

I. $\overline{\bar{z} + 3i} = z - 3i$

II. $|(2\bar{z} + 5)(\sqrt{2} - i)| = \sqrt{3} |2z + 5|$

III. $|2z + 3\bar{z}| \leq 4|\operatorname{Re}(z)| + |z|$ {Hint: Use triangular inequality}

$|z_1 \pm z_2| \leq |z_1| + |z_2|$ or $|z_1 \pm z_2| \geq ||z_1| - |z_2||$.

4. Find the modulus and argument of the following complex numbers:

(i) $\frac{2-i}{2+i}$

(ii) $\frac{\sqrt{3}+i}{\sqrt{3}-i}$

(iii) $\left(\frac{1 + \sqrt{3}i}{1 - \sqrt{3}i}\right)^2$

5. Prove that $|z-i| = |z+i|$ represents a straight line.

6. Prove that $|z+2i| + |z-2i| = 6$ represents an ellipse.

7. Find an equation of a circle center at (2,3) with radius 3.

8. Sketch the region in xy - plane represented by the following set of points:

$\operatorname{Re}(\bar{z} - 1) = 2$

$\operatorname{Im}(z^2) = 4$

$$\left| \frac{2z-3}{2z+3} \right| = 1$$

$$\operatorname{Re}(z) + \operatorname{Im}(z) = 0$$